

ASSIGNMENT 1

Marks: 50

Deadline: 03 June, 2026

BUS TICKET BOOKING & BILLING SYSTEM

OVERVIEW

Design and implement a Bus Ticket Booking & Billing System for a transportation company using C# and Object-Oriented Programming (OOP) principles. The application must enable users to book bus tickets, select seats, generate invoices, and process payments. Your implementation must demonstrate a thorough understanding of OOP concepts and appropriate application of SOLID design principles.

1. USER MANAGEMENT

The system must support the creation and management of user accounts.

Each user shall be assigned:

- A unique identifier (User ID)
- Full name
- Mobile number
- Email address

A single user may book multiple tickets. The system must maintain an accurate record of all user associated bookings.

2. BUS MANAGEMENT

The system must accommodate multiple buses within its fleet.

Each bus shall be configured with:

- A unique identifier (Bus ID)
- Coach number
- Bus classification (e.g., Business, Economy)
- Total seating capacity determined by bus classification

The system must accurately monitor and distinguish between available and reserved seats for each bus.

3. SCHEDULE MANAGEMENT

Each bus may operate on multiple schedules.

A schedule shall consist of:

- Departure city
- Arrival city
- Departure date and time
- Ticket price

Every schedule must be explicitly associated with a corresponding bus in the fleet.

4. TICKET BOOKING

The system must provide users with the capability to:

- Browse and select an available schedule
- Choose a preferred seat
- Complete a ticket reservation

The system shall enforce the following constraints:

- Validate seat numbers against the bus classification and its defined seating capacity
- Prevent duplicate reservations of the same seat for a given schedule

Upon successful completion of payment:

- A ticket must be generated and issued to the user
- The selected seat must be permanently marked as reserved

5. INVOICE & PAYMENT

Every confirmed ticket booking must automatically generate an invoice.

Each invoice shall contain:

- Invoice ID
- Ticket ID
- User ID
- Amount due
- Invoice generation date

- Payment status (Paid / Unpaid)

Users must have the ability to:

- Retrieve and view all their generated invoices
- Submit payment for any outstanding (unpaid) invoices

6. VIEWING & OPERATIONS

The system must expose the following functional operations:

- Create User
- Display All Users
- Create Bus
- Display All Buses
- Create Schedule
- Display All Schedules
- Display Schedule Details
- Book Ticket
- Display User Invoices
- Process Invoice Payment
- Display User Tickets

TECHNICAL REQUIREMENTS

- Implementation must be done in C#
- The application must be a Console Application
- Proper use of OOP pillars: Encapsulation, Inheritance, Polymorphism, and Abstraction
- Application of SOLID principles must be evident in the codebase structure and class design
- All entities must be modeled as classes with appropriate properties and methods

DELIVERABLES

Submit a complete, compilable, and runnable C# solution containing all source code files required to demonstrate the functionality described above.